THANDO CHIPANGO

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PROFESSIONAL SUMMARY

Information Technology student at the National College of Information Technology (NACIT), completing a Level 5 Diploma in Computing with Business Management (Levels 3 and 4 completed), combining full-stack development, data science and analytics, and cybersecurity with core business principles. Hands-on experience building and deploying web platforms, a network intrusion detection system and a poultry management system piloted with 10 farmers, developing databases in PHP, MySQL and MongoDB, and cleaning and analysing more than 66,000 records in Python and SQL. Experienced with REDCap, ODK, SurveyCTO and STATA. Passionate about Agritech, data, cybersecurity, mobile and full-stack development, and building secure, scalable systems that solve real-world problems.

EDUCATION

National College of Information Technology (NACIT) *Expected 2027*

Level 5 Diploma in Computing with Business Management (current)

- Relevant coursework: Backend Web Development, Big Data Analysis and Data Science, Mobile App Development, Principles of Business Operations.

Level 4 Diploma in Computing with Business Management (completed 2025)

- Relevant coursework: Frontend Web Development, Computer Networks, Databases, eBusiness, Understanding Business Organizations, Essentials of Business Management.

Level 3 Diploma in Computing (completed 2025)

- Relevant coursework: Python Programming, Web Technologies, Computer Science, Object Oriented Programming, System Analysis and Design.

DATA SCIENCE, DATA ANALYTICS & BIG DATA

- Data science and analytics: Python (pandas, NLTK, scikit-learn) in Jupyter and VS Code, and STATA; data cleaning, profiling and validation, descriptive statistics, frequency analysis, text processing and sentiment classification with model comparison.
- Big data and data analysis: Level 5 Big Data Analysis and Data Science coursework, and an end-to-end project on three linked datasets of more than 66,000 records, from loading and cleaning to analysis, reporting and a business strategy.
- Research data collection: REDCap, ODK and SurveyCTO electronic data collection tools.
- Databases and data quality: SQL (SQLite), MySQL, MongoDB and PHP; referential integrity checks, cross-checking results between tools, and clear documentation of every data change.

EXPERIENCE

Co-founder, Chief Strategy Officer & Web Developer, NexaCode Systems *2026*

Next.js 14, Tailwind CSS, JavaScript | Live: nexa-web-swart.vercel.app

- Lead company strategy and documentation for a six-person freelance team offering web development, mobile app development and cybersecurity services.
- Designed and built the company's primary marketing and business platform, architected on Next.js for performance and SEO and deployed on Vercel.
- Implemented a responsive Tailwind CSS design with a clear visual hierarchy for service offerings.
- Diagnosed and resolved a critical Content Security Policy (CSP) nonce mismatch that was causing sitewide hydration failures.
- Integrated Playwright end-to-end testing with an automated validation pipeline that runs before production deploys.
- Set out an improvement roadmap covering SEO, WCAG accessibility, security hardening and code modernization.

Systems & UX Audit Consultant, TaskMate Malawi (independent project) *2026*

- Audited a live MVP (taskmate-malawi-hub.lovable.app) across user experience, technical architecture and conversion metrics.
- Identified 6 high-priority barriers, including pricing ambiguity and friction in payment security flows, and proposed alternative UI solutions to build user trust.
- Developed a 4-phase implementation roadmap to improve mobile responsiveness (target 60-70% mobile traffic) and conversion rates.
- Recommended technical SEO improvements: schema.org structured data, heading hierarchies and XML sitemaps.

VOLUNTEER EXPERIENCE

Media & Data Volunteer, Theatre Renaissance Cabaret International Festival 2026 *Jul – Aug 2026*

Mwezi Arts, Lilongwe, Malawi

- Supported a 5-day international theatre festival with 11 participating countries, including Brazil, France, Tunisia, Zambia and Zimbabwe.
- Collected and curated media content (footage and data) from TV stations and individual contributors for multi-platform social media distribution.
- Worked with an international team during live operations, ensuring timely content delivery across all festival platforms.
- Managed on-site logistics, including accreditation and laptop-based equipment coordination, solving problems in real time.

PROJECTS

Poultry Management System | React, JavaScript, Tailwind CSS | Jul 2025 – Present

- Developing a web platform that helps smallholder farmers in Lilongwe track and manage poultry operations; launched as a pilot with 10 local farmers, using their feedback to improve the experience.
- Built a React dashboard for flock inventory (bird count, breeds, ages, health status) and a feed module tracking consumption, costs and stock levels.
- Created production tracking for daily egg collection, mortality rates and vaccination schedules, plus financial tracking to support data-driven decisions.
- Designed a mobile-responsive interface for smartphones and low-bandwidth connections.

Network Intrusion Detection System (NIDS) | Python, Scapy, Flask, Chart.js | Jan – Jun 2026 | github.com/Thando-rgb/NIDS

- Built from scratch a system that captures live network traffic with Scapy and detects port scans, SYN floods and ARP sweep reconnaissance in real time using sliding time-window thresholds.
- Added persistent timestamped logging and a live Flask and Chart.js dashboard with auto-refresh, colour-coded alerts, an attack-breakdown chart and an events table.
- Tested with a Kali Linux attacker VM against a Windows defender using nmap and hping3, confirming detection of all three attack types.
- Solved Windows interface naming differences, ARP versus ICMP behaviour on local networks and Npcap driver configuration.

Data Science and Big Data Project: Google Local Datasets | Python, pandas, NLTK, scikit-learn, SQLite | Oct 2026

- Loaded, structured and cleaned three datasets (22,476 places, 21,767 reviews, 21,858 users): fixed file-parsing errors, converted data types and split free-text fields with regular expressions.
- Profiled data quality (missing values, duplicates, broken links between tables) and left gaps unfilled rather than inventing values, documenting each decision.
- Processed review text with language detection, contraction expansion, spell correction, tokenisation and lemmatisation.
- Cross-checked pandas results with SQL queries in SQLite; the counts matched.
- Classified the sentiment of 15,960 English reviews, comparing VADER (83.4% accuracy), TextBlob (80.4%) and logistic regression (85.3%) on a held-out test set, then turned the findings into a business strategy and assessed data ethics and GDPR safeguards.

Cooking Website Database | PHP, MySQL, MongoDB

- Developed the database for a cooking website where users register, log in, add recipes and post reviews, with passwords stored securely (hashed or encrypted) for registration and login.

Personal Portfolio Website | HTML, CSS, JavaScript | 2025 | thando-portfolio-steel.vercel.app

- Developed a responsive, mobile-first portfolio with smooth-scroll animations, presenting my projects, technical skills and contact details in clear modular sections.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, SQL, PHP

Data & analytics: pandas, NLTK, scikit-learn, Jupyter, STATA, REDCap, ODK, SurveyCTO; MySQL, MongoDB, SQLite; data cleaning and profiling, descriptive statistics, sentiment analysis

Frameworks: React, React Native, Next.js, Flask, Tailwind CSS

Security & networking: Scapy, Kali Linux, network security, networking

Tools: Git, Linux, Node.js, VS Code, Playwright

Mobile development: Flutter, Swift

Professional development: pursuing certifications in Networking, Cloud Computing and Cybersecurity through Nzeru Hub

REFERENCES

Glory Jumbo, Lecturer, NACIT | jumboglory01@gmail.com

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